SSV Conclusions B

A concrete superfluid holographic screen undershoots Newton's constant by 38 orders: a mode-counting negative

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Abstract

This note records the sharpest quantitative result the Saturated Superfluid Vacuum (SSV) programme produced about emergent gravity, and it is a negative. Entropic and holographic gravity (Jacobson, Verlinde, Padmanabhan) keep the holographic screen *abstract*. SSV's distinctive move was to give the screen a concrete microscopic substrate — a logarithmic-Schrödinger condensate whose self-defined acoustic horizon carries an area-law entanglement entropy that can be *computed* rather than assumed. The form survives: the entanglement entropy is area-law, as expected for a local quadratic Hamiltonian. The magnitude does not. Counting the actual degrees of freedom the substrate offers per grain gives $\mathcal{O}(10)$ Bogoliubov modes — a few bits — against the $N \simeq 1.7 \times 10^{38}$ required to reproduce G. The screen, made literal and counted, misses Newton's constant by roughly 10^{37}. This is the Sakharov / Susskind–Uglum species problem, here *demonstrated by direct mode-counting* rather than argued, and it is a clean constraint on any reading that takes a holographic screen to be a real reservoir of local degrees of freedom. Per the repository's rule, the negative is stated at full strength.

1 The setup: making the screen literal

In the thermodynamic route to gravity, the field equation follows from $\delta Q = T\,dS$ applied to local causal horizons (Jacobson 1995), with the entropy taken to be area-law, $S = A/4\ell_P^2$. Verlinde's entropic-force reading and Padmanabhan's surface-thermodynamics share this skeleton. In all of them the holographic screen is a bookkeeping surface: one assumes an area-law entropy with the Planck-scale coefficient and does not ask what physical degrees of freedom supply it.

SSV's one genuinely distinctive contribution is to refuse that abstraction. The vacuum is a real medium with order parameter $\Psi = \sqrt{\rho}\,e^{i\theta}$ governed by a logarithmic Schrödinger equation; the screen is a self-defined acoustic horizon, the surface where the background flow turns sonic (Visser). Because the medium is concrete, two things that are normally assumed become *computable*: the area law, and the coefficient.

2 The form survives (and is the cheap half)

Quantising the LogSE Bogoliubov fluctuations and computing the vacuum entanglement entropy by the correlator method (Peschel; Casini–Huerta) via the Srednicki radial decomposition gives an area law. Across a static acoustic inhomogeneity (a "dumb hole") the scaling exponent tracks a provably area-law gapped-scalar calibrator and sits a full ~ 1.2 in log–log slope below a volume-law thermal control; the entropy-per-area is universal at the few-percent level; and the chiral-shear term is *exactly* silent (`instruments/paper_v/horizon_entanglement_entropy.py`).

This must be stated honestly (repository rule 1): the area law is the *expected* side. It follows from the LogSE quadratic Hamiltonian being *local* — generic for any local free field (Srednicki 1993; Bombelli–Koul–Lee–Sorkin 1986; Jacobson's entanglement-equilibrium, 2016). So the computation *earns* (turns assumed into derived-from-locality) the area law that the SSV gravity papers had merely asserted by cell-counting, but it is not a surprise and not a discovery. The discriminator was shown to be negative-capable — it correctly rejects volume-law data — which is what makes the next step trustworthy.

3 The magnitude fails (and is the whole point)

The coefficient is the test. Writing $\delta Q = T_H\, dS$ with the computed area-law $dS = \eta\, dA$ closes the Jacobson construction to $\nabla^2\Phi = 4\pi G_{\mathrm{eff}}\, e/c^2$ with $G_{\mathrm{eff}} = 1/(4\hbar c\,\eta)$ (`instruments/paper_v/dumb_hole_surfac`
To recover the measured G, the entropy density η must correspond to

$$N \;=\; \frac{1}{\alpha_G} \;=\; \frac{\hbar c}{G\, m_p^2} \;=\; \left(\frac{m_P}{m_p}\right)^2 \;\simeq\; 1.7\times 10^{38} \tag{1}$$

microstates per grain — this is just the Planck-to-proton hierarchy.

Now count what the substrate actually provides. The localised excitations of a LogSE vortex core — the bound Bogoliubov (BdG) modes — number $\mathcal{O}(10)$, i.e. a few bits, per healing-length cell (`instruments/paper_i/vortex_core_mode_spectrum.py`). The continuum modes are delocalised and do not localise entropy on the grain. So the concrete screen supplies of order 10^1 states where 10^{38} are needed: a shortfall of about 10^{37}, and correspondingly G_{eff} overshoots the Newtonian value by $(a_p/\ell_P)^2 \approx 1.7\times 10^{38}$.

4 What the negative establishes

When a holographic screen is given a real microscopic substrate and its localised degrees of freedom are *counted*, the area law emerges for free but the Planck-scale coefficient does not: the count is short by $\sim$38 orders of magnitude. A holographic screen that is a literal reservoir of local degrees of freedom cannot, in this construction, reproduce G. Newton's constant must enter below the effective theory's cutoff — as a trans-Planckian input, not an emergent output.

This is the Sakharov induced-gravity problem and the Susskind–Uglum species problem, which Volovik identifies as the structural weak point of superfluid gravity. The contribution here is not the existence of the gap but its *concrete demonstration*: rather than estimating an induced Newton constant from a cutoff, one builds the medium, computes its horizon entanglement entropy, counts its grain degrees of freedom, and reads off the shortfall directly. That makes the failure model-specific and checkable rather than a heuristic worry.

5 Caveats, stated plainly

- The identity $(a_p/\ell_P)^2 = 1/\alpha_G$ is a *tautology*: it restates the hierarchy and is not a derivation. It must not be read backwards as "deriving N." The negative is the *measured* count ($\mathcal{O}(10)$), confronted with this target.

- Bound modes are finite by construction; continuum modes are delocalised. A genuinely different substrate (e.g. a rank-2, ^{3}He-A-like order parameter with sub-grain structure) is not excluded by this argument — it simply is not the LogSE medium, and would have to supply its own 10^{38} states.
- The area-law computation assumes irrotational flow for the exact chiral silence; a rotational/vortex-sink horizon carries a bounded residual risk that was not closed.

None of these weaken the headline: in the medium actually proposed, the screen is $\sim$37 orders too sparse to be the source of G.

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